

Automotive Manufacture Report

1. **Problem Statement:** The automotive manufacturing company purchases different automotive parts from multiple suppliers. The company needs to monitor supplier Performance, delivery reliability, product quality, defects, shortages, lead, time and procurement costs.

The analysis helps identify supplier and manufacturing areas that require improvement and supports better operational decision – making.

2. **Business Requirements:**

- Analyze supplier performance.
- Identify delayed orders.
- Calculate on-time delivery percentage.
- Calculate shortage quantity.
- Analyze defective quantities.
- Calculate defective percentage.
- Identify suppliers with poor delivery performance.
- Identify suppliers with high defects and high lead time.
- Analyze procurement cost.
- Compare different categories.
- Analyze warehouse and regional performance.
- Identify areas for operational improvement.

3. **Tools Used:**

- MYSQL: Data Analysis and SQL queries
- Excel: Data Validation and Basic Calculations
- Power BI: Dashboard and Visualization
- Python: Data Cleaning and Exploratory Analysis

➤ **SQL concepts used**

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- Aggregate Functions
- INNER JOIN

- LEFT JOIN
- RIGHT JOIN
- Subqueries

4. Analysis:

The Analysis was performed on automotive manufacturing data containing information about orders, suppliers, quantities, delivery status, defects, lead time, cost, quality warehouse, and region.

➤ Supplier Analysis

- Number of orders handled by each supplier.
- Delayed orders by supplier.
- On_time delivery performance.
- Total ordered and delivered quantities
- Supplier – wise defect quantities
- Average lead time.

➤ Delivery Analysis

Calculated:

- Shortage Quantity = Ordered Quantity – Delivered Quantity
- Also Analyzed:
- On- time orders.
- Delayed Orders.
- On – Time Delivery Percentage.
- Suppliers with poor delivery performance.

➤ Quality Analysis

Calculated

- Defect Percentage = Defect Quantity / Delivered Quantity * 100
- Analyzed
- Total defective quantity
- Supplier-wise defects.
- Part-category defect
- Quality status
- Suppliers with high defects and high lead time.

➤ Cost Analysis

Analyzed:

- Total procurement cost
- Supplier – wise procurement cost

- Part – category cost
- High-cost suppliers and categories

➤ **Warehouse & Region Analysis**

- Compared
- Orders by warehouse
- Delivery Performance by warehouse
- Orders by region
- On-time percentage by region.

5. Insights

- Supplier delivery performance varies by supplier, with some suppliers having significantly more delayed orders.
- On-time delivery needs improvement, as a considerable number of orders are delayed.
- Shortage quantity exists between ordered and delivered quantities, indicating incomplete order fulfillment.
- Some suppliers have high defect quantities, requiring additional quality monitoring.
- Suppliers with both high defects and high lead time represent higher operational risk.
- Certain part categories contribute more to defects and procurement cost.
- Warehouse and regional analysis show that delivery performance is not consistent across locations.
- Supplier – level analysis can help the company decide which suppliers need performance improvement or closer monitoring.

6. Conclusion

The automotive manufacturing analysis provides a clear view of supplier, delivery, quality, cost, and operational performance.

The analysis helps identify delayed suppliers, defective parts, shortages, high lead-time suppliers, and high – cost categories.

By monitoring these KPIs regularly, the company can improve supplier performance, reduce defects and shortages, improve on – time delivery, and control procurement costs.

